Infinite Dyck Paths Algebra for Unbounded Interval Temporal Logic

Introduction

Three readings of a Dyck path

The subject of this paper is an interval temporal logic, and Dyck paths are the way we draw it. Following [1], take a finite linear order of time instants $t_1 < \ldots < t_{n-1}$, identified with the chain $[n-1] = \{1, \ldots, n-1\}$, and evaluate propositions on its intervals — the spans $[i, j]$ of consecutive instants — rather than on single instants. Among these propositions single out the *hereditary* ones, those whose truth on a span passes to every subspan: this is the logic of subintervals, the fragment of Halpern–Shoham logic [2] built on the single relation "during". A hereditary proposition is known once the intervals on which it holds are, and these form a family closed under taking subintervals, that is a down-set of the intervals of $[n-1]$ ordered by inclusion. Dyck paths enter as the geometric representation of such families, hence of the propositions themselves: an interval is drawn as a pyramid, a family of intervals as the skyline of its pyramids, and the pointwise order of two drawings is the entailment of the propositions they stand for. Everything below is meant to keep this reading available once the time line is no longer bounded.

The analysis of the finite Dyck lattices in [1] rests on one observation: a finite Dyck path can be read in three ways. Write D_n for the Dyck paths of semilength n ordered by geometric inclusion $P \leq Q$ when P lies weakly below Q. This is a finite distributive lattice, so by Birkhoff's representation theorem [3] it is the lattice of down-sets of its join-irreducible elements, and those are known: a join-irreducible is a path all of whose factors are hills except for a single pyramid $u^k d^k$ with $k \geq 2$, and it is determined by the interval of atoms lying below it. Writing $\mathrm{Int}([n-1])$ for the intervals $[i, j]$, $1 \leq i \leq j \leq n-1$, of the chain $[n-1]$ ordered by inclusion, this reads

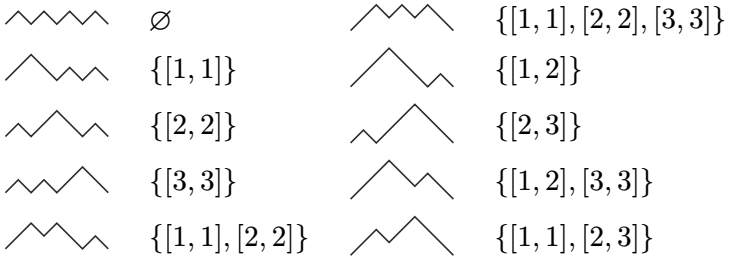
$$D_n \cong \mathcal{O}(\mathrm{Int}([n-1])),$$

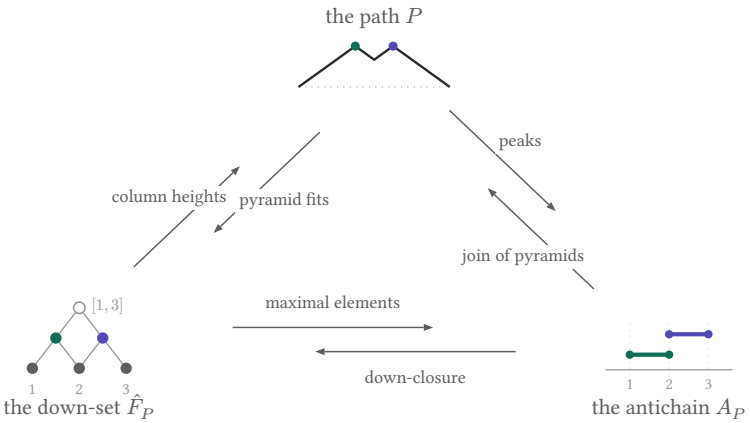
the lattice of down-sets of $\mathrm{Int}([n-1])$. A path, a down-set of intervals and the antichain of that down-set's maximal elements are three names for one object.

Example. Take $n = 4$, so the time line is the chain $[3] = \{1, 2, 3\}$ and $\mathrm{Int}([3])$ has the six elements $[1,1], [2,2], [3,3], [1,2], [2,3], [1,3]$. For $P = uuududdd$, with height profile $0, 1, 2, 3, 2, 3, 2, 1, 0$, the test $h_P(i + j) \geq j - i + 2$ runs over those six:

$$\begin{array}{llllll} [1,1] & h_P(2) = 2 \geq 2 & [2,2] & h_P(4) = 2 \geq 2 & [3,3] & h_P(6) = 2 \geq 2 \\ [1,2] & h_P(3) = 3 \geq 3 & [2,3] & h_P(5) = 3 \geq 3 & [1,3] & h_P(4) = 2 \not\geq 4 \end{array}$$

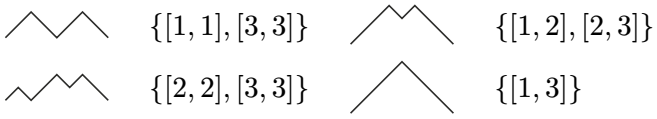
So $\hat{F}_P$ is everything but $[1,3]$, and its maximal elements are $A_P = \{[1,2], [2,3]\}$ — the same answer the peak rule gives at a glance. Running the computation over all of D_4 labels each of its fourteen paths with an antichain of intervals of $[3]$:





reading	on $P = uuududdd$	read off from
the path P		the coatom of D_4, with peaks above the axis at abscissas 3 and 5
the antichain A_P	$\{[1,2],[2,3]\}$	one maximal interval per peak of P above the axis
the down-set $\hat{F}_P$	$\mathrm{Int}([3]) \setminus \{[1,3]\}$	the down-closure of A_P

Figure 1: The three readings, on the coatom $P = uuududdd$ of D_4.



Reading the two columns against Figure 4: the empty antichain is $\bot_4$, the single interval $[1,3]$ spanning the whole chain is $\top_4$, and the number of intervals in A_P is the number of nontrivial peaks of P.

From a bounded time line to an unbounded one

Everything above is finite with the chain $[n-1]$: the intervals are the time spans of a bounded history, and the lattice they generate is bounded with them. The suggestion this paper follows is to let that chain run off to the right and keep the triangle standing, or as much of it as will stand. An unbounded Dyck path is still to be read as a down-set of intervals of the time line $\mathbb{N}_{\geq 1}$, hence as a hereditary proposition about an unbounded future, and the question becomes which connectives such paths support once no last stage closes them off. The passage between a path and its down-set survives intact; the antichain of maximal intervals does not always, since a down-set of an infinite poset may have no maximal elements at all, and we will keep the intervals but not the antichain.

What holds the triangle up is a representation theorem, and it is worth saying on which scale ours sits. Birkhoff's theorem is a statement about finite distributive lattices. Its general form for bounded distributive lattices is Priestley duality, which trades the finite poset of join-irreducibles for an ordered topological space; beside it stands the topology-free representation of the down-set lattice $\mathcal{O}(P)$ of an arbitrary poset P by its completely join-irreducible elements, the principal down-sets [3]. We will not need the strength of Priestley duality and do not use it, but our work follows this spectrum: the stages D_n are Birkhoff's case, and the ambient object in which we place their limit remains the down-set lattice of a poset, the finite intervals of $\mathbb{N}_{\geq 1}$, where the correspondence between the poset and the completely join-irreducible elements — now the paths made of a single pyramid — survives unchanged.

> Instead of considering a finite total order, one can consider either an infinite (but discrete) total order or a more general partial order. In the former case, the combinatorial structure that is

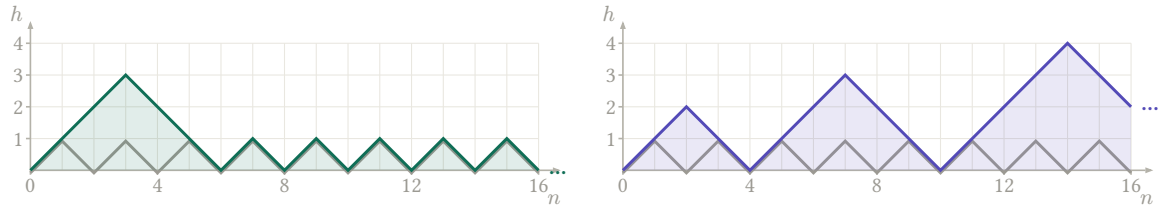
involved could be some notion of infinite Dyck path, that is, a Dyck path having infinite length. There are at least two different ways to formalize this suggestion: either working in the **set of all Dyck paths** (of any length), which is naturally partially ordered, and can be probably interpreted as some kind of infinitary limit object for the family of Dyck lattices; or introducing **Dyck paths having infinite support**, i.e., Dyck paths unbounded on the right which do not have a suffix consisting of the concatenation of an infinite number of peaks lying on the x-axis.

— Ferrari

Therefore, we denote:

- **Model 1:** the set of all Dyck paths;
- **Model 2:** the set of Dyck paths with infinite support.

Both models live inside the same picture. Read an infinite word over $\Sigma = \{u, d\}^\omega$ as a lattice path and compare it to the bottom element $\bot = (ud)^\omega$, the infinite concatenation of peaks lying on the axis: the two models are exactly the two ways a path can behave against $\bot$ in the long run.



Ambient definitions

For a path $P = p_0 p_1 p_2 \ldots$, we define the function $h_P(n)$ that returns the height of P after its first n steps.

$$h_P(0) = 0, \qquad h_P(n) = |\{i < n \mid p_i = u\}| - |\{i < n \mid p_i = d\}|.$$

$$\mathcal{D}_\omega = \{P = (p_i)_{i \in \mathbb{N}} \in \{u, d\}^\omega \mid \forall n \in \mathbb{N}, h_P(n) \geq 0\}$$

Definition 1 (geometric order)**.** For $P, Q \in \mathcal{D}_\omega$ set

$$P \leq Q \quad \Leftrightarrow \quad \forall n \in \mathbb{N}, h_P(n) \leq h_Q(n),$$

and say that P lies *weakly below* Q. We call $\leq$ the *geometric order*, since it compares the two drawings pointwise. It restricts to each D_n, and $(D_n, \leq)$ is the Dyck lattice of order n of [1].

We import the vocabulary of [1] about specific Dyck paths, transposed to infinite words. A *return* of a path P is an index $n > 0$ with P crossing the x axis i.e. $h_P(n) = 0$; a *peak* is a pair of consecutive steps $p_n p_{n+1} = ud$, and a *hill* is a peak at height 0, i.e. one with $h_P(n) = 0$. A *pyramid* is a block $u^k d^k$ ($k \geq 1$) starting and ending on the axis, so a hill is the pyramid with $k = 1$. Finally, a *factor* of P is a block of consecutive steps running between two consecutive returns (or from the origin to the first return); a hill is also called a *trivial factor*. In the finite case every Dyck path is the juxtaposition of its factors; here the same holds for the prefix of P up to its last return, and a path with infinitely many returns is the juxtaposition of infinitely many factors. The atom π_i are the path whose factors are all hills except for a single pyramid $uudd$.

Definition 2 (limit path)**.** In this language, we denote $\bot$ as the unique path all of whose factors are hills.

$$\bot = (ud)^\omega, \qquad h_\bot(n) = \begin{cases} 0 \text{ if } & n = 0 \bmod 2 \\ 1 \text{ if } & n = 1 \bmod 2 \end{cases}.$$

The *support* of a path measures how far it strays from the floor $\bot$: it collects the times at which P does not sit exactly where $\bot$ does. Since $\bot$ is nothing but hills, this amounts to recording where P does something other than a hill: a path of finite support is one whose factors are hills from some point on — a finite Dyck path followed by an infinite run of hills — whereas a path of infinite support keeps producing nontrivial factors arbitrarily late. This is precisely the line separating Model 1 from Model 2.

$$\mathrm{supp}(P) = \{n \in \mathbb{N} \mid h_P(n) \neq h_\bot(n)\}, \qquad \mathrm{supp}_{\mathrm{at}}(P) = \{i \geq 1 \mid h_P(2i) > 0\}.$$

The second variant samples only the even times, and there $\bot$ is on the axis: since paths never go below 0, "differing from $\bot$" and "being strictly above the axis" say the same thing. The next two lemmas show that this sampling loses nothing, so we may use whichever variant is convenient.

Lemma 3. $\mathrm{supp}_{\mathrm{at}}(P) = \{i \mid 2i \in \mathrm{supp}(P)\}$.

Lemma 4. $|\mathrm{supp}(P)| < \infty \Leftrightarrow |\mathrm{supp}_{\mathrm{at}}(P)| < \infty$.

Lemma 5. *$i \in \mathrm{supp}_{\mathrm{at}}(P) \Leftrightarrow \pi_i \leq P$, where $\pi_i = (ud)^{i-1}uudd(ud)^\omega$.*

Ferrari represents a path of D_n by an antichain of intervals of $[n-1]$, namely the orders of the atoms lying below each join-irreducible in its decomposition [1]. We keep the intervals, but not the antichain.

Definition 6 (interval representation)**.** Let Int denote the set of finite nonempty intervals $I = [a, b]$ of $\mathbb{N}_{\geq 1}$, ordered by inclusion. For $I = [a, b] \in \mathrm{Int}$ put

$$J_I = (ud)^{a-1}u^{b-a+2}d^{b-a+2}(ud)^\omega,$$

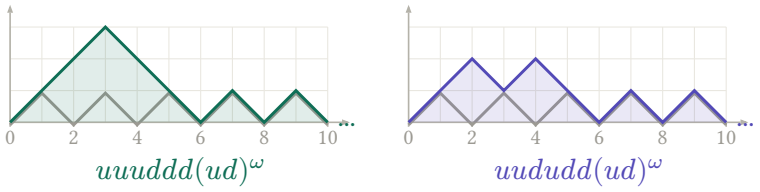
the path all of whose factors are hills except for a single pyramid, and set

$$\hat{F}_P = \{I \in \mathrm{Int} \mid J_I \leq P\}.$$

Each J_I is join-irreducible in $\mathcal{D}_\omega$, and $J_{[i,i]} = \pi_i$; hence $\mathrm{supp}_{\mathrm{at}}(P) = \{i \mid [i, i] \in \hat{F}_P\}$: the support records exactly the singletons of $\hat{F}_P$, and forgets every wider interval.

Lemma 7. *$J_I \leq J_{I'}$ if and only if $I \subseteq I'$. In particular $\hat{F}_P$ is a down-set of* Int.

Example. Knowing $\mathrm{supp}_{\mathrm{at}}(P)$ is not enough to know P. Both paths below have $\mathrm{supp}_{\mathrm{at}} = \{1, 2\}$, yet $uududd(ud)^\omega < uuuddd(ud)^\omega$.



$uuuddd(ud)^\omega$ $\qquad$ $uududd(ud)^\omega$

The order is exactly what the picture shows: at every abscissa the right path sits weakly below the left one, and that pointwise comparison of heights is all that $\leq$ means.

The two are nevertheless different paths. The left one is a single pyramid, hence join-irreducible with atom-interval $[1, 2]$, while the right one is $\pi_1 \vee \pi_2$; what separates them is the down-set, $\hat{F} = \{[1, 1], [2, 2], [1, 2]\}$ against $\hat{F} = \{[1, 1], [2, 2]\}$, of which $\mathrm{supp}_{\mathrm{at}}(P)$ retains only the singletons. So $\mathrm{supp}_{\mathrm{at}}(P)$ tells us whether P has finite support, but not which path P is; only $\hat{F}_P$ does that.

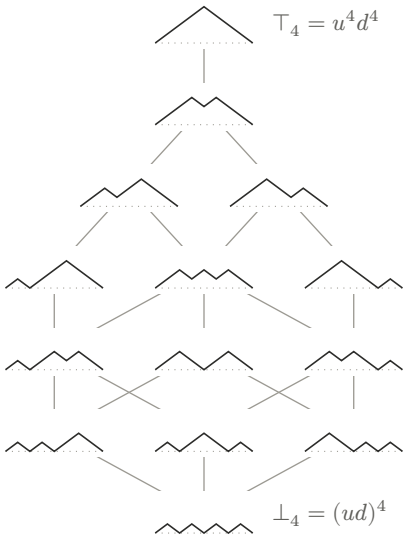


Figure 4: The Dyck lattice D_4, drawn in the geometric order.

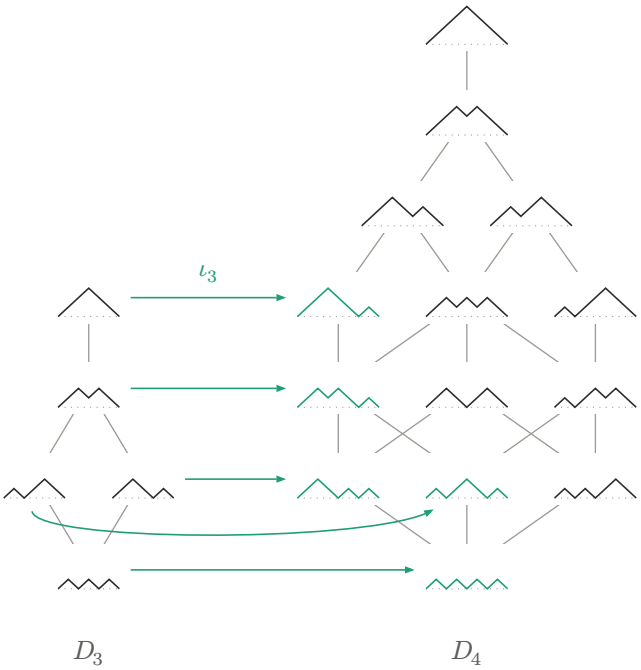


Figure 5: The case $\iota_3 : D_3 \hookrightarrow D_4$.

Lemma 8. *Pointwise* min *and* sup *of height functions of paths in $\mathcal{D}_\omega$ are height functions of paths in $\mathcal{D}_\omega$; hence $(\mathcal{D}_\omega, \leq)$ is a complete lattice with $\bot = (ud)^\omega$, $\top = u^\omega$, and binary meets distribute over arbitrary joins, so $\mathcal{D}_\omega$ is a complete Heyting algebra.*

Dyck paths lattice

Each $(D_n, \leq)$ is a finite distributive lattice. The Figure 4 illustrate the lattice for $n = 4$. The bottom is the path of four hills $\bot_4$, the top the single pyramid $\top_4$, and an edge joins two paths when one covers the other in the geometric order.

These finite lattices are the stages of a single tower. To pass from one stage to the next we leave the path alone and append a hill at the right end by the inclusion function ι defined below. The Figure 5 draws the case ι_3.

$$\iota_n : D_n \to D_{n+1}, \qquad \iota_n(P) = Pud.$$

Appending a hill is also the extension that fixes the floor, sending $\bot_n = (ud)^n$ to $\bot_{n+1}$, which keeps the whole tower compatible with the reference path against which support is measured. In the interval representation the map does nothing at all: the intervals representing P are unchanged and only the ambient index set grows from $[n-1]$ to $[n]$, the new atom π_n failing to lie below Pud because Pud has returned to the axis at abscissa $2n$. Appending hills lets a path of one stage be compared with a path of any later one, and two paths should count as equal when the tower eventually makes them so.

Definition 9 (eventual equality)**.** For $n \le k$ let $\iota_{n,k} : D_n \to D_k$ denote the composite $\iota_{k-1} \circ \cdots \circ \iota_n$, with $\iota_{n,n} = \mathrm{id}$; that is, $\iota_{n,k}(P) = P(ud)^{k-n}$. On the disjoint union

$$\coprod_{n\in\mathbb{N}} D_n = \{(n, P) \mid n \in \mathbb{N}, P \in D_n\}$$

define

$$(n, P) \sim (m, Q) \quad \Leftrightarrow \quad \exists k \ge \max(n, m) : \iota_{n,k}(P) = \iota_{m,k}(Q).$$

The relation $\sim$ is obviously an equivalence relation. Its classes are what let paths of different lengths be compared at all: $P \in D_n$ and $Q \in D_m$ with $n \neq m$.

intro D_ω lattice and figure Figure 6

Model 1

We restate that Model 1 corresponds to the set of all Dyck paths (of any length), which is naturally partially ordered, and can probably be interpreted as some kind of infinitary limit object for the family of Dyck lattices [4], [5], [6].

Lemma 10. *ι_n is injective.*

Lemma 11. *ι_n is an order-embedding preserving function.*

The infinitary limit object $\mathcal{D}_{\mathrm{colim}}$ for the family of Dyck lattices, denoted $\operatorname{colim} D_n$, is the colimit of the chain $D_0 \xrightarrow{\iota} D_1 \xrightarrow{\iota} D_2 \to \cdots$ along the maps ι_n: the smallest object in which every stage sits compatibly. Therefore, the $\mathcal{D}_{\mathrm{colim}}$ is simply the increasing union of the tower quotiented by $\sim$.

Definition 12 (Direct limit of D_n)**.** Writing $[(n, P)]$ for the $\sim$-class of (n, P), put

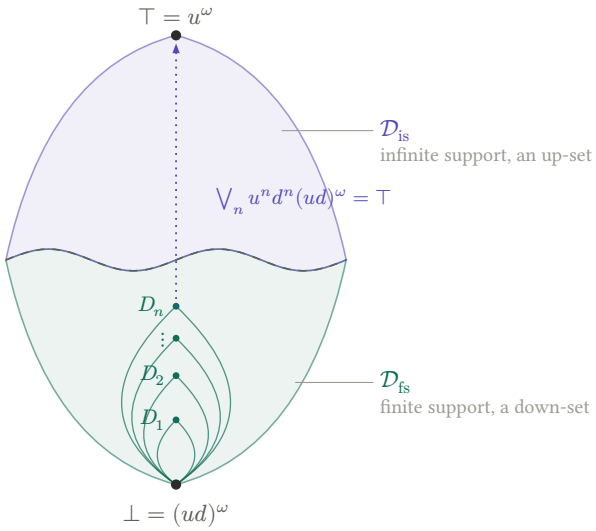


Figure 6: The two strata of $\mathcal{D}_\omega$, and the tower $D_1 \subset D_2 \subset \cdots D_\omega$.

$$\mathcal{D}_{\mathrm{colim}} = \left(\coprod_{n \in \mathbb{N}} D_n \right) / \sim, \qquad c_n : D_n \to \mathcal{D}_{\mathrm{colim}}, \quad c_n(P) = [(n, P)],$$

and equip $\mathcal{D}_{\mathrm{colim}}$ with

$$[(n,P)] \le [(m,Q)] \quad \Leftrightarrow \quad \exists k \ge \max(n,m) : \iota_{n,k}(P) \underset{D_k}{\le} \iota_{m,k}(Q),$$

$$[(n,P)] \vee [(m,Q)] = \left[\left(k, \iota_{n,k}(P) \vee_{D_k} \iota_{m,k}(Q) \right) \right],$$

and

$$[(n,P)] \wedge [(m,Q)] = \left[\left(k, \iota_{n,k}(P) \wedge_{D_k} \iota_{m,k}(Q) \right) \right]$$

for any $k \ge \max(n,m)$, and $\bot = [(0, \varepsilon)]$ with ε the empty path.

An element of $\mathcal{D}_{\mathrm{colim}}$ is thus a finite Dyck path together with the convention that the flat ground to its right does not count. Letting that ground run forever is exactly a subset of $\mathcal{D}_\omega$.

$$j : \mathcal{D}_{\mathrm{colim}} \hookrightarrow \mathcal{D}_\omega \qquad j([(n,P)]) = P(ud)^\omega.$$

Proposition 13. *j is an order-embedding, and $j(\mathcal{D}_{\mathrm{colim}}) = \{P \in \mathcal{D}_\omega \mid |\mathrm{supp}(P)| < \infty\}$.*

Algebraic representation

Model 1 has a second possible description, it is an initial algebra. Recall that an algebra for an endofunctor F is just a map $F(X) \to X$ and a coalgebra is a map $X \to F(X)$; neither carries the associativity that an algebra over a monad, or a coalgebra over a comonad, would supply. Least and greatest fixpoints of F are written μF and νF respectively.

Definition 14 (initial algebra)**.** An *F-algebra* is a pair (X, α) with $\alpha : F(X) \to X$, and a morphism of F-algebras $(X,\alpha) \to (Y,\beta)$ is a map $f : X \to Y$ with $f \circ \alpha = \beta \circ F(f)$. An algebra $(\mu F, \mathrm{in})$ is *initial* when for every F-algebra (X,α) there is exactly one such f, that is a unique

$$\mathrm{fold}(\alpha) : \mu F \to X \quad \text{with} \quad \mathrm{fold}(\alpha) \circ \mathrm{in} = \alpha \circ F(\mathrm{fold}(\alpha)).$$

cite

Recall finite Dyck words are constructed with the grammar $S ::= \varepsilon \mid uSdS$ generates, and that grammar is a signature functor.

Definition 15 (Dyck initial algebra)**.** Let $F : \mathrm{Set} \to \mathrm{Set}$ be

$$F(X) = 1 + X \times X,$$

with constructors ε (i.e. $\varnothing$) for the terminal object 1 and $u(-)d(-)$ for the product $X \times X$, the latter reading a pair (A,B) as the word $uAdB$.

Proposition 16. *F is finitary, so its initial algebra exists and is computed by the initial chain $\varnothing \to F(\varnothing) \to F^2(\varnothing) \to \cdots$, which converges at ω:*

$$\mu F = \bigcup_{k \in \mathbb{N}} F^k(\varnothing),$$

with $\mathrm{in} : F(\mu F) \to \mu F$ *the pair of constructors. Its elements are the finite Dyck words before any quotient.*

Let $\sigma : F(\mathbb{N}) \to \mathbb{N}$ be the F-algebra $\sigma(*) = 0$ and $\sigma(a,b) = a + b + 1$, and set

n	terms of μF	word	count
0	ε	ε	$C_0 = 1$
1	$u\,\varepsilon\,d\,\varepsilon$	ud	$C_1 = 1$
2	$u\,\varepsilon\,d\,(u\,\varepsilon\,d\,\varepsilon)$	$udud$	$C_2 = 2$
	$u\,(u\,\varepsilon\,d\,\varepsilon)\,d\,\varepsilon$	$uudd$	
3	$u\,\varepsilon\,d\,(u\,\varepsilon\,d\,(u\,\varepsilon\,d\,\varepsilon))$	$ududud$	$C_3 = 5$
	$u\,\varepsilon\,d\,(u\,(u\,\varepsilon\,d\,\varepsilon)\,d\,\varepsilon)$	$uduudd$	
	$u\,(u\,\varepsilon\,d\,\varepsilon)\,d\,(u\,\varepsilon\,d\,\varepsilon)$	$uuddud$	
	$u\,(u\,\varepsilon\,d\,(u\,\varepsilon\,d\,\varepsilon))\,d\,\varepsilon$	$uududd$	
	$u\,(u\,(u\,\varepsilon\,d\,\varepsilon)\,d\,\varepsilon)\,d\,\varepsilon$	$uuuddd$	

Table 1: The correspondance of μF and Catalan number $C_n = \frac{1}{n+1}\binom{2n}{n}$

$$\ell := \mathrm{fold}(\sigma) : \mu F \to \mathbb{N}, \quad \ell(\varepsilon) = 0, \quad \ell(uAdB) = \ell(A) + \ell(B) + 1,$$

the unique map counting the uses of the constructor $u(-)d(-)$, i.e. the semilength.

Proposition 17. *The fibres of ℓ are the strata of the tower: $\ell^{-1}(n)$ is exactly the carrier of D_n, so that*

$$\mu F = \coprod_{n\in\mathbb{N}} \ell^{-1}(n) \cong \coprod_{n\in\mathbb{N}} D_n,$$

and $\left|\ell^{-1}(n)\right| = C_n$:

$$1, 1, 2, 5, 14, 42, 132, \ldots$$

the Catalan numbers.

define D_ω by the quotienting of μF

Model 2

$$\mathcal{D}_{\infty\text{-supp}} = \{P \in \mathcal{D}_\omega \mid |\mathrm{supp}(P)| = \infty\}, \qquad \mathcal{D}_\omega = j(\mathcal{D}_{\mathrm{colim}}) \sqcup \mathcal{D}_{\infty\text{-supp}}.$$

Unbounded Interval Temporal Logics

$$\mathcal{D}_{\mathrm{fs}} := \{P \in \mathcal{D}_\omega \mid |\mathrm{supp}(P)| < \infty\}, \qquad \mathcal{D}_{\mathrm{is}} := \{P \in \mathcal{D}_\omega \mid |\mathrm{supp}(P)| = \infty\}.$$

$$\mathcal{D}_\omega = \mathcal{D}_{\mathrm{fs}} \sqcup \mathcal{D}_{\mathrm{is}}.$$

Logic of Model 1

$$\mathcal{D}_1 := \operatorname*{colim}_n D_n \cong \mathcal{D}_{\mathrm{fs}} = \{P \in \mathcal{D}_\omega \mid |\mathrm{supp}(P)| < \infty\}.$$

$$\bot = (ud)^\omega \in \mathcal{D}_1, \qquad \top \notin \mathcal{D}_1.$$

$$P, Q \in \mathcal{D}_1 \quad \Rightarrow \quad P \wedge Q \in \mathcal{D}_1, \qquad P \vee Q \in \mathcal{D}_1.$$

$$h_{P\wedge Q}(n) = \min\big(h_P(n), h_Q(n)\big), \qquad h_{P\vee Q}(n) = \max\big(h_P(n), h_Q(n)\big).$$

$$\mathrm{supp}(P \wedge Q) = \mathrm{supp}(P) \cap \mathrm{supp}(Q), \qquad \mathrm{supp}(P \vee Q) = \mathrm{supp}(P) \cup \mathrm{supp}(Q).$$

$$\mathcal{D}_1 = \text{distributive lattice with bottom, without top.}$$

$$\neg\exists\top \quad \Rightarrow \quad \neg\exists \text{ global Heyting implication.}$$

$$\mathcal{L}_1 = \{\bot, \wedge, \vee\}.$$

Logic of Model 2

$$\mathcal{D}_2 \coloneqq \mathcal{D}_{\text{is}} = \{P \in \mathcal{D}_\omega \mid |\text{supp}(P)| = \infty\}.$$

$$(ud)^\omega \notin \mathcal{D}_2, \qquad \top = u^\omega \in \mathcal{D}_2.$$

$$P, Q \in \mathcal{D}_2 \quad \Rightarrow \quad P \vee Q \in \mathcal{D}_2.$$

$$\text{supp}(P \vee Q) = \text{supp}(P) \cup \text{supp}(Q), \qquad |\text{supp}(P)| = \infty \quad \Rightarrow \quad |\text{supp}(P \vee Q)| = \infty.$$

$$P, Q \in \mathcal{D}_2 \quad \not\Rightarrow \quad P \wedge Q \in \mathcal{D}_2.$$

$$\text{supp}(P \wedge Q) = \text{supp}(P) \cap \text{supp}(Q), \qquad |\text{supp}(P)| = |\text{supp}(Q)| = \infty \quad \not\Rightarrow \quad |\text{supp}(P) \cap \text{supp}(Q)| = \infty.$$

$$\bot \notin \mathcal{D}_2, \qquad \wedge \text{ is not closed.}$$

$$\mathcal{D}_2 = \text{join-semilattice with top}, \qquad \mathcal{L}_2 = \{\top, \vee\}.$$

$$\neg\exists\bot, \qquad \neg\exists \text{ global } \wedge, \qquad \neg\exists \text{ global } \to, \qquad \neg\exists \text{ global negation.}$$

Completion

$$\mathcal{D}_\omega = \mathcal{D}_{\text{fs}} \sqcup \mathcal{D}_{\text{is}}, \qquad \mathcal{D}_1 \to \mathcal{D}_\omega \leftarrow \mathcal{D}_2.$$

$$\mathcal{D}_1 \cong \mathcal{D}_{\text{fs}} \subseteq \mathcal{D}_\omega, \qquad \mathcal{D}_2 = \mathcal{D}_{\text{is}} \subseteq \mathcal{D}_\omega.$$

$$\bot = (ud)^\omega, \qquad \top = u^\omega.$$

$$h_{\sup_{i\in I} P_i}(n) = \sup_{i\in I} h_{P_i}(n), \qquad h_{\inf_{i\in I} P_i}(n) = \inf_{i\in I} h_{P_i}(n).$$

$$P \wedge \left(\sup_{i\in I} Q_i\right) = \sup_{i\in I}(P \wedge Q_i).$$

$$\mathcal{D}_\omega = \text{frame} = \text{complete Heyting algebra.}$$

$$P \to Q = \sup\{R \in \mathcal{D}_\omega \mid P \wedge R \leq Q\}, \qquad \neg P = P \to \bot.$$

$$\mathcal{L}_\omega = \{\bot, \top, \wedge, \vee, \to, \neg\} + \{\sup_{i\in I}, \inf_{i\in I}\}.$$

$$\text{Idl}(\mathcal{D}_1) \cong \mathcal{D}_\omega.$$

Final overview

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